Stable-Baselines3 Docs - Reliable Reinforcement Learning Implementations

Stable Baselines3 (SB3) is a set of reliable implementations of reinforcement learning algorithms in PyTorch. It is the next major version of Stable Baselines.

GitHub repository: https://github.com/DLR-RM/stable-baselines3

Paper: https://jmlr.org/papers/volume22/20-1364/20-1364.pdf

RL Baselines3 Zoo (training framework for SB3): https://github.com/DLR-RM/rl-baselines3-zoo

RL Baselines3 Zoo provides a collection of pre-trained agents, scripts for training, evaluating agents, tuning hyperparameters, plotting results, and recording videos.

SB3 Contrib (experimental RL code, latest algorithms): https://github.com/Stable-Baselines-Team/stable-baselines3-contrib

SBX (SB3 + Jax): https://github.com/araffin/sbx

Main Features

- Unified structure for all algorithms
- PEP 8 compliant (unified code style)
- Documented functions and classes
- Tests, high code coverage, and type hints
- Clean code
- TensorBoard support
- **The performance of each algorithm was tested** (see *Results* section in their respective page)

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- Installation
  - Prerequisites
  - Bleeding-edge version
  - Development version
  - Using Docker Images
- Getting Started
- Reinforcement Learning Tips and Tricks
  - General advice when using Reinforcement Learning
  - Which algorithm should I use?
  - Tips and Tricks when creating a custom environment
  - Tips and Tricks when implementing an RL algorithm
- Reinforcement Learning Resources
- RL Algorithms
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  - Training exceeds `total_timesteps`
- Examples
  - Try it online with Colab Notebooks!
  - Basic Usage: Training, Saving, Loading
  - Multiprocessing: Unleashing the Power of Vectorized Environments
  - Multiprocessing with off-policy algorithms
  - Dict Observations
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  - Callbacks: Evaluate Agent Performance
  - Atari Games
  - PyBullet: Normalizing input features
  - Hindsight Experience Replay (HER)
  - Learning Rate Schedule
  - Advanced Saving and Loading
  - Accessing and modifying model parameters
  - SB3 with Isaac Lab, Brax, Procgen, EnvPool
  - SB3 with DeepMind Control (dm_control)
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    - Modifying Vectorized Environments Attributes
    - Vectorized Environments Wrappers
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    - DummyVecEnv
    - SubprocVecEnv
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    - Logging Figures/Plots
    - Logging Videos
    - Logging Hyperparameters
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    - Hugging Face 🤗
    - MLFLow
- RL Baselines3 Zoo
    - Installation

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    - Colab Notebook: Try it Online!
- SB3 Contrib
    - Why create this repository?
- Stable Baselines Jax (SBX)
- Plotting
    - Recommended Approach: RL Baselines3 Zoo Plotting
    - Real-Time Monitoring
    - Monitor File Format
    - Basic SB3 Plotting (Simple Use Cases)
- Imitation Learning
- Migrating from Stable-Baselines
    - Overview
    - How to migrate?
    - Breaking Changes
    - New Features (SB3 vs SB2)
- Dealing with NaNs and infs
    - How and why?
    - Anomaly detection with PyTorch
    - Numpy parameters
    - VecCheckNan Wrapper
    - RL Model hyperparameters
    - Missing values from datasets
- Developer Guide
    - Algorithms Structure
    - Where to start?
    - Pre-Processing
    - Policy Structure
    - Probability distributions
    - State-Dependent Exploration
    - Misc
- On saving and loading

    - Zip-archive
- Exporting models
    - Background
    - Export to ONNX
    - Trace/Export to C++
    - Export to ONNX-JS / ONNX Runtime Web
    - Export to TensorFlow.js
    - Export to TFLite / Coral (Edge TPU)
    - Manual export
    - SBX (SB3 + Jax) Export

RL Algorithms

- Base RL Class
- A2C
- DDPG
- DQN
- HER
- PPO
- SAC
- TD3

Common

- Atari Wrappers
- Environments Utils
- Custom Environments
- Probability Distributions
- Evaluation Helper
- Gym Environment Checker
- Monitor Wrapper
- Logger
- Action Noise
- Utils

Misc

- Changelog

- Projects

Citing Stable Baselines3

To cite this project in publications:

```
@article{stable-baselines3,
  author  = {Antonin Raffin and Ashley Hill and Adam Gleave and Anssi Kanervisto
and Maximilian Ernestus and Noah Dormann},
  title   = {Stable-Baselines3: Reliable Reinforcement Learning Implementations},
  journal = {Journal of Machine Learning Research},
  year    = {2021},
  volume  = {22},
  number  = {268},
  pages   = {1-8},
  url     = {https://jmlr.org/papers/v22/20-1364.html}
}
```

Note: If you need to refer to a specific version of SB3, you can also use the Zenodo DOI.

Contributing

For anyone interested in making the RL baselines better, there are still some improvements that need to be done. You can check issues in the repository.

If you want to contribute, please read CONTRIBUTING.md first.

Indices and tables

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